Bar-chain models of E_1-chiral quantum groups and the E_N ladder

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April 2026

1 The Yangian as E_1-chiral quantum group

The Yangian $Y_\hbar(\mathfrak{sl}_2)$ is presented by generators $t_{ij}^{(r)}$, $1 \le i, j \le 2$, $r \ge 1$, assembled into the generating series $T(u) = \mathbf{1} + \sum_{r\ge 1} T^{(r)} u^{-r}$ with $T^{(r)} = (t_{ij}^{(r)})$, subject to the RTT relation

$$R(u-v)\, T_1(u)\, T_2(v) = T_2(v)\, T_1(u)\, R(u-v), \qquad R(z) = 1 - \frac{\hbar\, P}{z}, \tag{1}$$

where $P \in \mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2)$ is the permutation.

Multiplication. The RTT relation (1) defines a $\mathbb{Z}$-filtered associative algebra. It is *not* commutative: $[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj} t_{il}^{(1)} - \delta_{il} t_{kj}^{(1)})$. Multiplication is E_1 and no more.

Coproduct. The spectral coproduct

$$\Delta_z \colon Y_\hbar(\mathfrak{sl}_2) \longrightarrow Y_\hbar(\mathfrak{sl}_2) \,\widehat{\otimes}\, Y_\hbar(\mathfrak{sl}_2), \qquad \Delta_z(T(u)) = T(u) \otimes T(u-z), \tag{2}$$

is coassociative: $(\mathrm{id} \widehat{\otimes} \Delta_w) \circ \Delta_z = (\Delta_z \widehat{\otimes} \mathrm{id}) \circ \Delta_{z+w}$. At $z = 0$ it reduces to the Drinfeld coproduct $\Delta_0(T(u)) = T(u) \otimes T(u)$.

R-matrix and quasi-triangularity. The Yang R-matrix $R(z) = 1 - \hbar P/z$ satisfies the Yang–Baxter equation

$$R_{12}(z_1 - z_2)\, R_{13}(z_1 - z_3)\, R_{23}(z_2 - z_3) = R_{23}(z_2 - z_3)\, R_{13}(z_1 - z_3)\, R_{12}(z_1 - z_2). \tag{3}$$

Quasi-triangularity: $R(z)\, \Delta_z(a) = \Delta_z^{\mathrm{op}}(a)\, R(z)$ for all $a \in Y_\hbar(\mathfrak{sl}_2)$.

Braided module category. For $Y_\hbar(\mathfrak{sl}_2)$-modules M, N, the spectral coproduct (2) defines a fusion product $M \otimes_z N$ by pulling back the $(Y_\hbar \widehat{\otimes} Y_\hbar)$-action. The braiding

$$\beta_{M,N}(z) \colon M \otimes_z N \xrightarrow{R(z)\cdot\sigma} N \otimes_z M$$

satisfies the braid relation by (3).

What makes $Y_\hbar(\mathfrak{sl}_2)$ genuinely E_1. The RTT relation (1) is *not* derived from a local OPE on a curve. There is no vertex algebra V for which $Y_\hbar(\mathfrak{sl}_2)$ arises as a chiral envelope. The ordering of tensor factors

is independent input. In the hierarchy

$$E_1\text{-chiral} \supsetneq E_\infty\text{-chiral (vertex algebras)},$$

the Yangian sits in $E_1 \setminus E_\infty$: genuinely nonlocal.

This example earns the following definition.

2 E_1-chiral quantum groups

Definition 2.1 (Ordered Koszul chiral algebra; (Vol I ??)). *An E_1-chiral algebra $\mathcal{A}$ on a smooth curve X is* ordered Koszul *if (i) its ordered bar complex $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ over the pure braid Orlik–Solomon operad* $\mathrm{OS}(\mathrm{PBr}_n)$ *is quadratic-Koszul in the Shelton–Yuzvinsky sense; (ii) the induced A_∞-structure $\{m_k^{\mathrm{ord}}\}_{k\geq 2}$ on $\mathcal{A}$ satisfies the Stasheff boundary-of-associahedra relations; (iii) the R-twisted Σ_n-descent identifying $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ with the symmetric-bar shadow $\overline{B}^{\Sigma}(\mathcal{A})$ holds at the level of chain complexes.*

Proposition 2.2 (Yangian as ordered-Koszul E_1-chiral algebra; (Vol I ??)). *The Yangian $Y_\hbar(\mathfrak{g})^{\mathrm{ch}}$ in Drinfeld's second presentation, extended to the $\mathcal{D}$-module on a smooth curve X by the factorisation-algebra construction of* [?CostelloWittenYamazaki]*, is an ordered Koszul chiral algebra in the sense of Definition 2.1. Equivalently (Etingof–Kazhdan 2005 + this monograph's $\overline{B}^{\mathrm{ord}}$ construction): the pure braid Orlik–Solomon Koszulity produces a flat A_∞ structure whose symmetrisation recovers the Etingof–Kazhdan quantum vertex algebra on the Koszul locus.*

Remark 2.3 (Why the Yangian is the archetypal E_1-chiral input). The five notions of E_1-chiral algebra (strict ChirAss, A_∞ in $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$, Etingof–Kazhdan quantum VA, A_∞ in E_1-chiral, factorization on $\mathrm{Ran}^{\mathrm{ord}}(X)$) coincide on the ordered Koszul locus where the Drinfeld associator acts freely, through Proposition 2.2. The Yangian sits in the complement of E_∞-chiral (standard vertex algebras) and is genuinely nonlocal.

Definition 2.4 (E_1-chiral quantum group). *An E_1-**chiral quantum group** on a smooth curve X is a tuple $(\mathcal{A}, \mu, \Delta, R, S)$ where:*

- *(i) $\mathcal{A}$ is an E_1-chiral algebra on X: an E_1-algebra object in factorisation algebras on X;*
- *(ii) μ is the E_1-chiral multiplication (the A_∞-structure $\{m_k\}_{k\geq 2}$ encoding the ordered OPE);*
- *(iii) $\Delta\colon \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$ is a chiral coproduct (coassociative coalgebra structure on $\mathcal{A}$ itself, in the chiral tensor category);*
- *(iv) $R(z) \in \mathcal{A}^{\widehat{\otimes}_2}$ is a spectral R-matrix satisfying the Yang–Baxter equation* (3)*;*
- *(v) $R(z)\,\Delta(a) = \Delta^{\mathrm{op}}(a)\,R(z)$ for all $a \in \mathcal{A}$ (quasi-triangularity);*
- *(vi) $S\colon \mathcal{A} \to \mathcal{A}$ is an antipode satisfying $\mu \circ (\mathrm{id} \otimes S) \circ \Delta = \eta \circ \varepsilon$ and its mirror.*

The coproduct Δ is structure on $\mathcal{A}$. It is not the deconcatenation coproduct on the bar complex $B(\mathcal{A})$: deconcatenation is a structural feature of any bar construction and carries no information about $\mathcal{A}$ beyond what the product already encodes.

Theorem 2.5 (Categorical level shift). *Let $(\mathcal{A}, \mu, \Delta, R, S)$ be an E_1-chiral quantum group. Then:*

(a) *$\mathcal{A}$ is an E_1-algebra.*

(b) $\mathrm{Mod}_{\mathcal{A}}$ *is an* E_2*-algebra object in* Cat: *it is a braided monoidal category with braiding* $\beta_{M,N}(z) = R(z) \cdot \sigma$.

(c) *The derived chiral centre* $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) := \mathrm{ChirHoch}^\bullet(\mathcal{A}, \mathcal{A})$ *carries an* E_3*-chiral algebra structure via the chiral higher Deligne theorem (Vol II* ??*). On cohomology this is* E_2*; the* E_3 *upgrade at chain level is the content of the pentagon edges* (3) ↔ (4) ↔ (5) *of the* $\mathsf{SC}^{\mathrm{ch,top}}$*-heptagon.*

E_3 *lives on* $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$*, not on* $\mathcal{A}$.

Proof. (a) is the definition.

(b) The coproduct Δ defines the monoidal product $M \otimes_\Delta N$ on $\mathrm{Mod}_{\mathcal{A}}$ by restriction of scalars along Δ. Coassociativity of Δ gives associativity of $\otimes_\Delta$. The R-matrix $R(z)$ and quasi-triangularity give the braiding natural transformation $\beta_{M,N} \colon M \otimes N \to N \otimes M$ satisfying the hexagon axioms (which follow from the YBE (3)). The antipode S gives duals.

(c) The chiral higher Deligne theorem (Vol II ??): for an E_1-chiral algebra $\mathcal{A}$ on a smooth curve X, the chiral Hochschild cochains $\mathrm{ChirHoch}^\bullet(\mathcal{A}, \mathcal{A})$ carry a canonical E_3-chiral structure delivered by the pentagon edges (3) ↔ (4) ↔ (5) of the $\mathsf{SC}^{\mathrm{ch,top}}$-heptagon (Vol II `ch:sc-chtop-heptagon`). On cohomology this reduces to the Gerstenhaber E_2 of the classical chiral Deligne conjecture; the E_3 upgrade is the present chapter's content. Associator discipline the chain-level E_3-action depends on a choice of Drinfeld associator, while the cohomological action is associator-free. □

3 The bar complex: what it sees and what it does not

Let $\mathcal{A}$ be an E_1-chiral algebra on a smooth curve X.

Proposition 3.1 (Bar complex as E_1 coalgebra [1, Theorem A])**.** *The bar construction*

$$B(\mathcal{A}) = \bigoplus_{n \geq 0} (s\mathcal{A})^{\otimes n}, \qquad d_B = \sum_{k \geq 2} \sum_i m_k^{(i)},$$

$$\Delta(a_1 | \cdots | a_n) = \sum_{i=0}^{n} (a_1 | \cdots | a_i) \otimes (a_{i+1} | \cdots | a_n).$$

is an E_1 *dg coassociative coalgebra on* $\mathrm{Ran}(X)$*. The differential* d_B *is a coderivation of* Δ*. The cobar construction* Ω *provides a left adjoint:* $\Omega B(\mathcal{A}) \simeq \mathcal{A}$ *for augmented* $\mathcal{A}$*.*

The bar complex computes $B(\mathcal{A}) \simeq k \otimes^{\mathbf{L}}_{\mathcal{A}} k$, the factorisation homology of the interval $[0, 1]$ with augmentation boundary conditions. It is not factorisation homology of the line: $\int_{\mathbb{R}} \mathcal{A} \simeq \mathcal{A}$.

Proposition 3.2 (Classical r-matrix from collision residues)**.** *Let* $\mathcal{A} = V_k(\mathfrak{g})$ *be the Kac–Moody vertex algebra at level* k*. The genus-0, degree-2 collision residue of the bar differential gives*

$$r(z) = \frac{k\,\Omega}{z},$$

where $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$ *is the Casimir. This* $r(z)$ *satisfies the classical Yang–Baxter equation.*

Proof. The m_2 component of d_B encodes the singular OPE $J^a(z)J^b(w) \sim k\,\delta^{ab}/(z-w) + f^{ab}_c J^c(w)/(z-w)$ (in the orthonormal basis where the Killing form is δ^{ab}). The $d\log$-extraction procedure replaces the OPE pole $1/(z-w)$ by the logarithmic form $d\log(z-w)$. The collision residue

at $z = w$ in degree 2 extracts the coefficient $k\,\Omega = k \sum_a J^a \otimes J_a$, divided by the spectral parameter $z = z_1 - z_2$. The CYBE follows from the Jacobi identity of $\mathfrak{g}$ (equivalently, from the codimension-2 stratum of $\mathrm{FM}_3(\mathbb{C})$, which imposes the three-term relation). □

Proposition 3.3 (Curvature). *At genus $g \geq 1$, the bar complex has*

$$d_B^2 = \kappa_{\mathrm{ch}} \cdot \omega_g, \tag{4}$$

where κ_{ch} is the shadow obstruction class and ω_g is the Arakelov $(1,1)$-form on the genus-g surface, normalised so that $\int_{\Sigma_g} \omega_g = 1$.

The bar complex sees the *product* (fusion, encoded in d_B) and the *classical r-matrix* (from collision residues).

It does *not* see:

(i) the chiral coproduct $\Delta\colon \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$ (the coproduct encodes splitting, not fusion; it is independent algebraic structure);

(ii) the full quantum R-matrix $R(z)$ (only its classical limit $r(z)$ appears in the shadow tower);

(iii) quasi-triangularity or the antipode.

The shadow obstruction tower $\{F_g\}_{g\geq 1}$, extracted from (4), detects collision data. The coproduct encodes the opposite operation and is invisible to the tower.

Sources of the chiral coproduct: Gaiotto–Rapčák–Zhou [4] construct coproducts on deformed double current algebras of $\mathfrak{gl}_K$ from M2–M5 fusions; Jindal–Kaubrys–Latyntsev [5] construct vertex coproducts on CoHAs of quivers with potential, recovering Drinfeld's deformed coproduct for ADE quivers.

4 The E_N ladder

For an E_1-chiral algebra $\mathcal{A}$ on X, the following structures are obtained by ascending the E_N hierarchy. Each step adds one E_1 factor via Dunn additivity.

E_1: the boundary algebra

$\mathcal{A}$ is E_1-chiral: an E_1-algebra in factorisation algebras on X. The bar complex $B(\mathcal{A})$ is the E_1 dg coassociative coalgebra of Proposition 3.1.

Status: proved (Proposition 3.1).

$E_1 \to E_2$: Hochschild and Deligne

Theorem 4.1 (E_2 on the derived centre). *The chiral derived centre*

$$Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}) = C_{\mathrm{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})$$

carries an E_2-chiral algebra structure. The E_2 decomposition $E_2 \simeq E_1^{\mathrm{hol}} \otimes E_1^{\mathrm{top}}$ (Dunn additivity) gives:

(a) *E_1^{hol}: the chiral algebra structure of $Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$ on X;*

(b) *E_1^{top}: the A_∞-algebra structure in the transverse direction.*

The pair $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\ \mathcal{A})$ *is the Swiss-cheese datum: the bulk* E_2 *acts on the boundary* E_1.

Status: proved. The chiral Deligne conjecture follows from Kontsevich formality and homotopy transfer to the chiral setting. Dunn additivity applies to $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, not to $\mathcal{A}$.

$E_2 \to E_3$-chiral: chiral higher Deligne

The E_3-chiral structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{ChirHoch}^\bullet(\mathcal{A}, \mathcal{A})$ is delivered unconditionally on the Koszul locus by the chiral higher Deligne theorem (Vol II ??), via the pentagon edges (3) $\leftrightarrow$ (4) $\leftrightarrow$ (5) of the $\mathsf{SC}^{\mathrm{ch,top}}$-heptagon. For chiral algebras of gauge-theoretic origin, this E_3-chiral structure is realised geometrically by the factorisation algebra of a 3d holomorphic-topological theory on $X \times \mathbb{R}$.

Theorem 4.2 (Chiral higher Deligne; (Vol II ??)). *Let* $\mathcal{A}$ *be an* E_1*-chiral algebra on a smooth complex curve* X *in the Koszul locus. Then* $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ *is the universal* E_3*-chiral algebra acted on by* $\mathsf{SC}^{\mathrm{ch,top}}$ *via the heptagon. Theorem H concentration* $\mathrm{ChirHoch}^{\geq 3}(\mathcal{A}) = 0$ *is the* consequence *of* E_3*-rigidity-at-a-point plus chiral PBW collapse, not a hypothesis.*

Theorem 4.3 (E_3-chiral realisation for Kac–Moody [7];). *For* $\mathcal{A} = V_k(\mathfrak{g})$ *at non-critical level* $k \neq -h^\vee$*, holomorphic Chern–Simons theory on* $X \times \mathbb{R}$ *provides the 3d HT bulk. The factorisation algebra of observables on* $X \times \mathbb{R}$ *is an* E_3*-chiral algebra, and its restriction to the boundary* $X \times \{0\}$ *recovers* $V_k(\mathfrak{g})$*. This geometric realisation agrees with the chiral higher Deligne* E_3*-chiral structure of Theorem 4.2 on cohomology; the chain-level agreement is associator-dependent above.*

Scope (AP-TOPOLOGIZATION). Kac–Moody proved by Costello–Li and Costello–Francis–Gwilliam. For $\mathcal{W}$-algebras via DS reduction, the bulk comes from DS boundary conditions and the BRST identity $T_{\mathrm{DS}} = [Q_{\mathrm{tot}}, G'_f]$ holds for any nilpotent f (the improvement term uses only Cartan currents). For general freely-generated conformal VAs, Khan–Zeng's 3d Poisson sigma model supplies the bulk. The chiral higher Deligne route gives the E_3-chiral structure unconditionally on the Koszul locus.

E_3-chiral $\to$ E_3-topological: topologisation

Theorem 4.4 (Topologisation for Kac–Moody;). *Let* $\mathcal{A} = V_k(\mathfrak{g})$ *at non-critical level* $k \neq -h^\vee$*. Write* $T(z) = \{Q, G(z)\}$ *where* $T(z)$ *is the Sugawara conformal vector,* Q *is the BRST operator, and* $G(z)$ *is the antighost. Then:*

(a) *Translations along* X *are* Q*-exact on the chain-level antighost refinement of Vol I ??, so the complex structure of* X *is irrelevant in the chain-level BRST cohomology.*

(b) *The* E_3*-chiral structure on* $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ *(Theorem 4.2, Theorem 4.3) descends to an* E_3*-topological structure on the original chain-level complex (class L via Dunn-additivity of* F^{CS}_k *on* $\mathbb{R} \times X$*).*

(c) *The resulting* E_3*-topological theory is Chern–Simons at level* $k + h^\vee$*.*

Theorem 4.5 (Topologisation tower, all classes; (Vol I ??, Vol II ??)). *On the original complex:*

(i) *class G (Heisenberg)* E_3^{top} *via Dunn on the commutative filtration;*

(ii) *class L (*$V_k(\mathfrak{g})$*,* $k \neq -h^\vee$*)* E_3^{top} *via Dunn on* F^{CS}_k *with explicit* $\eta_1^{(i)}, \eta_1^{(ii)}$ *making* $[Q, \tilde{G}_1] = T_{\mathrm{Sug}}$ *strict chain-level;*

(iii) *class C (*$\beta\gamma$*,* bc*)* E_3^{top} *via FMS bosonisation reducing to class G.*

In the weight-completed category:

(iv) *class M (Virasoro, $\mathcal{W}_N$) E_3^{top} via half-BRST and the MC5 weight-completed pattern (?? + ??).*

At critical level $k = -h^\vee$, Sugawara degenerates and the derived centre stops at E_2^{top} (dimension drop, not failure).
Higher-spin extension (Vol II ??): k inner stress tensors yield E_{k+2}^{top} via Dunn; for $\mathcal{W}_N$ this gives E_{N+1}^{top}; for $\mathcal{W}_\infty$ this is the Platonic endpoint E_∞^{top}.

Conjecture 4.6 (Topologisation for general conformal chiral algebras, original complex class M; AP–TOPOLOGIZATION). *Let $\mathcal{A}$ be a class-M chiral algebra at non-critical level. The weight-completed E_3^{top} structure of Theorem 4.5(iv) lifts to an E_3^{top} structure on the original complex (not merely weight-completed). Open for Virasoro and principal $\mathcal{W}_N$ at generic central charge on the original complex; BP (minimal $\mathcal{W}^{(2)}(\mathfrak{sl}_3)$) has a strict chain-level original-complex proof via branched-cover integralisation plus Galois descent (Vol II ??), indicating the route for general class M is branched-cover reduction to class L upstairs.*

At the *critical level* $k = -h^\vee$, the Sugawara construction degenerates. No non-critical conformal vector is available, so topologisation fails: the derived center remains E_3-chiral, but not E_3-topological.

Remark 4.7 (The generic case is $\mathrm{SC}^{\mathrm{ch,top}}$, not E_3). E_3-topological requires *both* a conformal vector at non-critical level *and* a 3d holomorphic-topological bulk theory. Chiral algebras lacking these inputs--critical-level Kac–Moody $V_{-h^\vee}(\mathfrak{g})$, E_1-chiral algebras (Yangians), Calabi–Yau functor outputs--carry the generic first-class structure $\mathrm{SC}^{\mathrm{ch,top}}$, in which the holomorphic and topological colours remain distinct.

The Heisenberg algebra $\mathcal{H}_k$ ($k \neq 0$) and lattice VOAs are *not* stuck at $\mathrm{SC}^{\mathrm{ch,top}}$: $\mathcal{H}_k$ carries the abelian Sugawara conformal vector $T = \frac{1}{2k}{:}JJ{:}$ ($c = 1$) and abelian holomorphic Chern–Simons theory provides the 3d HT bulk, so $Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{H}_k)$ reaches E_3-topological (the same proof as for non-abelian Kac–Moody, Theorem 4.4).

5 The modular operad: constructed; curved-Dunn: closed

The formerly conjectural chiral-A_∞ modular operad of the $E_2 \to E_3$ step has been constructed, and curved Dunn additivity has been closed at all genera. We record the statements.

Theorem 5.1 (Chiral-A_∞ modular operad; (Vol II `ch:modular-swiss-cheese-operad`)). *The modular operad $\mathcal{O}^{\text{ch-}A_\infty}$ exists and its algebras consist of:*

(a) *an E_1-chiral algebra structure on X (governed by $\mathsf{SC}^{\mathrm{ch,top}}$ on within-surface strata);*
(b) *an A_∞-algebra structure in the transverse direction $\mathbb{R}$;*
(c) *compatibility between transverse A_∞-operations and the E_1 ordering on X;*
(d) *at genus $g \geq 1$: curvature $d_{\mathrm{fib}}^2 = \kappa_{\mathrm{ch}}(\mathcal{A}) \cdot \omega_g$ as the bar intrinsic obstruction to extending the genus-0 structure to higher genera.*

The genus-0 part of $\mathcal{O}^{\text{ch-}A_\infty}$ is $\mathsf{SC}^{\mathrm{ch,top}} \times E_1^{\mathrm{tr}}$.

Theorem 5.2 (Curved Dunn additivity, all genera; (Vol II ??)). *At genus $g \geq 1$, the curved modular bar complex of $\mathcal{O}^{\text{ch-}A_\infty}$ factorises as a twisted Eilenberg–Zilber tensor product*

$$\mathbf{B}_{\mathrm{mod}}(\mathcal{O}^{\text{ch-}A_\infty}) \;\simeq\; \mathbf{B}_{\mathrm{mod}}(\mathsf{SC}^{\mathrm{ch,top}}) \;\otimes^{\tau}\; \mathbf{B}(E_1^{\mathrm{tr}}),$$

with twisting cochain τ built recursively from R-matrix monodromy at each genus-raising clutching. At genus 1, uncurving is explicit via Gauss–Manin on $\Lambda^2 H^1$ and Arakelov pairing on λ_1 (Vol II ??); at $g \geq 2$, vanishing of the curved-Dunn obstruction $H^2(\mathrm{TCo}^\bullet_g(\mathcal{A}), d_0) = 0$ transports from the modular-bootstrap complex via the explicit bridge map of Vol II ??.

The unified chiral quantum group

The nine families of chiral quantum group (Yangian $Y_\hbar(\mathfrak{g})$, affine Yangian $Y_\hbar(\hat{\mathfrak{g}})$, shifted Yangian $Y_\mu(\mathfrak{g})$, truncated shifted $Y_\mu^\lambda(\mathfrak{g})$ = BFN Coulomb branch, finite W-algebra $W^{\mathrm{fin}}(\mathfrak{g}, f)$, principal affine $W_k(\mathfrak{g}, f_{\mathrm{prin}})$, non-principal affine $W_k(\mathfrak{g}, f)$, Bershadsky–Polyakov $W_k(\mathfrak{sl}_3, f_{\mathrm{min}})$, orthogonal coideal Q^θ) are unified in a single four-parameter family $Q_{\mathfrak{g}}^{k,f,\mu}$ (Vol II `ch:unified-chiral-quantum-group`), with uniform Koszul, bar, and R-matrix data.

Theorem 5.3 (Chiral QG equivalence, ordered Koszul locus; (Vol I ??)). *On the ordered Koszul locus (Definition 2.1), the three structures -- classical r-matrix $r(z)$, A_∞ chiral algebra $\{m_k^{\mathrm{ord}}\}$, and ordered chiral coproduct Δ_z -- determine each other up to Drinfeld associator. Concretely, the Yangian triangle $\alpha \circ \gamma \circ \beta = \mathrm{id}$ is verified explicitly at order $\hbar$ for $Y_\hbar(\mathfrak{sl}_2)$. The equivalence covers E_1-chiral input (Yangian) as well as E_∞-chiral input (vertex algebras on the Koszul locus). The elliptic case is proved via Felder's dynamical $R(z, \tau)$ and Fay trisecant replacing the pentagon; the toroidal case is proved on the formal disk via Schiffmann–Vasserot CoHA and Miki DIM.*

Theorem 5.4 ($\mathfrak{gl}_N$ chiral QG, unconditional all $N \geq 2$; (Vol I ??)). *The principal $\mathcal{W}_N$ is reconstructed as the intersection of Feigin–Frenkel screening kernels inside a rank-$(N-1)$ Heisenberg,*

$$\mathcal{W}_N \;=\; \bigcap_{i=1}^{N-1} \ker Q_{\alpha_i} \;\subset\; \Pi_{\mathrm{FF}},$$

and the Drinfeld coproduct descends from the abelian Heisenberg parent to $\mathcal{W}_N$; the Molev qdet-central lemma is inscribed internally (Molev's antisymmetriser on the defining representation, not via external citation JKL26). The $\Psi = 0$ degenerate case produces the free abelian chiral algebra with identity QG; the $\Psi = 1$ free-boson point produces an OPE obstruction which vanishes by the screening-intersection compatibility. The two-parameter $\mathfrak{gl}_N$ RTT (bilinears $B_{\mathrm{tr}}, B_{\mathrm{ab}}$) matches $\mathrm{Sym}^2(\mathfrak{gl}_N^*)^{\mathfrak{gl}_N} = 2$-dimensional. Non-principal $\mathcal{W}$-hook types (nilpotent rank $r \leq N-3$) via parabolic screenings (Kac–Roan–Wakimoto, Arakawa).

Theorem 5.5 (Class-M weight-completed chiral QG; (Vol I ??)). *For $\mathcal{W}_{1+\infty}$ at self-dual level $c = 1$ and, more generally, for every class-M chiral algebra in the standard landscape at generic central charge, the chiral quantum group triangle (multiplication, coproduct, R-matrix) closes in the weight-completed category. The direct-sum category class-M triangle is obstructed at weight 4 by nonvanishing $S_4(\mathrm{Vir}_c) = 10/[c(5c+22)]$; this is the correct scope of the class-M chain-level statement.*

Theorem 5.6 (Spectral–categorical bridge; (Vol I ??)). *The E_2-Gerstenhaber braiding on $\mathrm{ChirHoch}^\bullet(\mathcal{A}, \mathcal{A})$ from chiral higher Deligne is associator-dependent: trivial on H^0, homotopic on cochains. The categorical braiding is recovered as the $\hbar$-expansion of the R-matrix braiding under the GRT_1 torsor action of Drinfeld associators. Position-dependence on $\mathrm{Mod}_{\mathcal{A}}$ arises from the $\mathcal{D}$-module structure via the KZB realisation of the spectral parameter.*

Theorem 5.7 (Class-M global triangle, weight-completed; (Vol II ??)). *For class-M $\mathcal{A}$ at generic central charge,*

$$\mathcal{A}_{\mathrm{bulk}} \;\simeq\; Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}_{\mathrm{boundary}}) \;\simeq\; \mathrm{HH}^*_{\mathrm{ch}}(\mathcal{A}^!_{\mathrm{line}})$$

holds in the weight-completed category (Vol I ?? + ??). At chain level on the original complex, the DS–Hochschild bridge (Vol II ??) reduces the class-M identification to the class-L identification for principal and hook-type Drinfeld–Sokolov reductions (Virasoro, principal $\mathcal{W}_N$, hook-type $\mathcal{W}_k(\mathfrak{g}, f)$).

6 The equivalence triangle for $\mathcal{W}_{1+\infty}$

The preceding sections establish the E_1-chiral quantum group definition (Definition 2.4) and note that the coproduct Δ is not visible in the shadow tower. For the Yangian $Y_\hbar(\mathfrak{sl}_2)$, all three data (multiplication, coproduct, R-matrix) are supplied by the RTT formalism. The class M version of the same package is Theorem 5.5: the triangle closes in the weight-completed category, and fails at the direct-sum level at weight 4 by $S_4(\mathrm{Vir}_c) = 10/[c(5c+22)]$. The present section records the explicit triangle functors $\alpha, \beta, \gamma^{-1}$ for $\mathcal{W}_{1+\infty}$ at self-dual level $c = 1$; all statements are understood in the weight-completed category unless noted.

The test case is $\mathcal{W}_{1+\infty}$ at the self-dual level $c = 1$, where the Maulik–Okounkov (MO) R-matrix $R^{\mathrm{MO}}(u)$ is proved [1, 3]. Strategy 3: use R^{MO} as input and reconstruct the remaining data (A_∞ operations and coproduct) via the *equivalence triangle*

$$\begin{tikzcd} & A_\infty\text{-structure } \{m_k\} \arrow[dr, "\beta"] & \\ R\text{-matrix } R(z) \arrow[ur, "\alpha"] \arrow[rr, "\gamma^{-1}=\beta\circ\alpha"'] & & \text{coproduct } \Delta \end{tikzcd} \tag{5}$$

whose three functors are:

(a) α: *R-matrix $\to A_\infty$.* Restrict the spectral R-matrix data from $\mathrm{FM}_n^{\mathrm{ord}}(\mathbb{C})$ to the Stasheff associahedra $K_n \hookrightarrow \mathrm{FM}_n^{\mathrm{ord}}(\mathbb{C})$. The boundary strata of K_n encode the A_∞ operations m_k: K_n is the Stasheff compactification of the moduli of n points on $\mathbb{R}$ (a codimension-0 submanifold of $\mathrm{FM}_n^{\mathrm{ord}}(\mathbb{C})$), and the boundary stratification $\partial K_n = \bigcup_{p+q=n+1} K_p \times K_q$ recovers the A_∞ associativity relations $\sum m_p \circ m_q = 0$.

(b) β: *$A_\infty \to$ coproduct.* Given the A_∞ operations $\{m_k\}$, form the ordered bar coalgebra $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with bar differential $d_B = \sum m_k^{(i)}$. The cobar construction $\Omega(B^{\mathrm{ord}}(\mathcal{A}))$ produces an A_∞ algebra; dualizing the deconcatenation coproduct on B^{ord} through Koszul duality gives the twisted coproduct $\Delta_z : \mathcal{A}^!_{\mathrm{line}} \to \mathcal{A}^!_{\mathrm{line}} \widehat{\otimes} \mathcal{A}^!_{\mathrm{line}}$ on the open-colour Koszul dual.

(c) $\gamma^{-1} = \beta \circ \alpha$: *$R$-matrix $\to$ coproduct* (composite). The MO R-matrix $R^{\mathrm{MO}}(u)$ produces the coproduct directly, bypassing the intermediate A_∞ structure.

Step α: extracting A_∞ from the MO R-matrix

The structure function of $\mathcal{W}_{1+\infty} \simeq Y(\widehat{\mathfrak{gl}}_1)$ is

$$g(u) = \frac{(u-h_1)(u-h_2)(u-h_3)}{(u+h_1)(u+h_2)(u+h_3)}, \qquad h_1 + h_2 + h_3 = 0. \tag{6}$$

The MO R-matrix on the charge-n Fock space $\mathcal{F}_n$ has diagonal eigenvalues $R_{\lambda\lambda}(u) = \prod_{s\in\lambda} g(u + c(s))$ where $c(s) = h_1 i + h_2 j$ is the content of box $s = (i, j)$ in partition λ [3].

The functor α restricts the $\mathrm{FM}_n(\mathbb{C})$ configuration-space data to associahedra:

Construction 6.1 (α: R-matrix to A_∞)**.** *At degree n, the A_∞ operation m_n^{ch} is extracted from the R-matrix as*

$$m_n^{\mathrm{ch}}(a_1, \dots, a_n) \;=\; \int_{K_n} \omega_n^R(z_1, \dots, z_n; a_1, \dots, a_n),$$

where ω_n^R is the propagator form on K_n twisted by the R-matrix:

$$\omega_n^R \;=\; \bigwedge_{i<j} d\log(z_i - z_j) \;\cdot\; R_{i,i+1}(z_i - z_{i+1}) \;\cdot\; a_1(z_1) \otimes \cdots \otimes a_n(z_n).$$

For $n = 2$: K_2 is a point, and m_2 is the binary product $m_2(a, b) = \mathrm{Res}_{z=0}\; a(z)\, b(0)\, d\log(z)$, the standard OPE extraction.

For $n = 3$: $K_3 \simeq [0, 1]$ is an interval. The integral over K_3 extracts the first higher product m_3^{ch} from the triple collision with the R-matrix twist:

$$m_3^{\mathrm{ch}}(a_1, a_2, a_3) \;=\; \int_0^1 \Big[\mathrm{Res}_{z_{12}=0}\; R_{12}(z_{12})\, a_1(z_1)\, a_2(z_2)\Big] \;\cdot\; a_3(z_3)\; d\log(z_{23})\, dt,$$

where the integration variable t parametrises K_3 and z_{12}, z_{23} are the successive differences. For class L (Kac–Moody, $R(z) = 1 + \hbar\, P/z$), the integral evaluates to zero: $m_3 = 0$, consistent with Koszulness.

For $\mathcal{W}_{1+\infty}$ (class M): $m_3^{\mathrm{ch}} \neq 0$. The structure function (6) contributes terms at every order in $1/z$, producing a nonvanishing K_3 integral. In the spin-2 channel (Virasoro at $c = 1$ inside $\mathcal{W}_{1+\infty}$), the m_3 operation encodes the first higher correction to the Virasoro OPE, consistent with $d_{\mathrm{gen}}(\mathrm{Vir}) = 3$ (generating depth 3: m_3 generates all higher m_k by recursive composition).

Step β: cobar dualization to the coproduct

Construction 6.2 (β: A_∞ to coproduct)**.** *Given the A_∞ operations $\{m_k^{\mathrm{ch}}\}$ from Construction 6.1, form the ordered bar complex*

$$B^{\mathrm{ord}}(\mathcal{W}_{1+\infty}) \;=\; T^c(s^{-1}\overline{\mathcal{W}_{1+\infty}})$$

with bar differential $d_B = \sum_{k\geq 2} m_k^{(i)}$ and deconcatenation coproduct. The open-colour Koszul dual is

$$\mathcal{A}^!_{\mathrm{line}} \;=\; H^*\big(B^{\mathrm{ord}}(\mathcal{W}_{1+\infty})\big)^\vee,$$

which is $Y(\widehat{\mathfrak{gl}}_1)$ (the affine Yangian of $\mathfrak{gl}_1$). The twisted coproduct on $\mathcal{A}^!_{\mathrm{line}}$ is obtained by dualizing the deconcatenation:

$$\Delta_z(f)(\bar{a}_1 \otimes \cdots \otimes \bar{a}_n) \;=\; \sum_{p=0}^{n} f(\bar{a}_1, \dots, \bar{a}_p) \;\otimes_z\; f(\bar{a}_{p+1}, \dots, \bar{a}_n),$$

where $\otimes_z$ denotes the completed tensor product with spectral parameter z tracking the position of the cut. The twisting by z arises from the holomorphic coordinate: cutting the ordered sequence at position p separates the configuration into two clusters whose relative position is z.

The composite γ^{-1}: direct R-matrix to coproduct

Proposition 6.3 (Cohomological coproduct from MO R-matrix)**.** *At the cohomological level, the composite $\gamma^{-1} = \beta \circ \alpha$ applied to the MO R-matrix $R^{\mathrm{MO}}(u)$ of (6) produces a well-defined coproduct*

$$\Delta_z^{\mathrm{MO}} \colon Y(\widehat{\mathfrak{gl}}_1) \longrightarrow Y(\widehat{\mathfrak{gl}}_1) \,\widehat{\otimes}\, Y(\widehat{\mathfrak{gl}}_1) \tag{7}$$

satisfying:

(i) Coassociativity at cohomology: $(\Delta_z \widehat{\otimes}\, \mathrm{id}) \circ \Delta_{z+w} = (\mathrm{id} \widehat{\otimes}\, \Delta_w) \circ \Delta_z$.

(ii) Quasi-triangularity at cohomology: $R^{\mathrm{MO}}(z)\, \Delta_z(a) = \Delta_z^{\mathrm{op}}(a)\, R^{\mathrm{MO}}(z)$ *for all* $a \in Y(\widehat{\mathfrak{gl}}_1)$.

(iii) Low-spin explicit form: *on the Cartan generators ψ_k (spin 1):*

$$\Delta_z(\psi(u)) \;=\; \psi(u) \otimes \psi(u-z),$$

where $\psi(u) = 1 + \sum_{k\geq 0} \psi_k\, u^{-k-1}$; on the positive root generators e_k (spin 2):

$$\Delta_z(e(u)) \;=\; e(u) \otimes 1 \;+\; \psi(u) \otimes e(u-z);$$

and on the negative root generators f_k:

$$\Delta_z(f(u)) \;=\; 1 \otimes f(u) \;+\; f(u-z) \otimes \psi(u).$$

These match the Schiffmann–Vasserot spectral coproduct at $z = 0$ and the Drinfeld new coproduct Δ_{new} at leading spectral order.

Proof sketch. The well-definedness at cohomology follows from the general principle: the bar-cobar adjunction $\Omega \circ B \simeq \mathrm{id}$ holds at the level of the *coderived category* for all shadow classes, including class M [1, thm:bv-bar-coderived]. The obstruction to a chain-level quasi-isomorphism (which exists for classes G, L, C but fails for class M) does not affect the cohomological coproduct, because:

(a) The MO R-matrix is a cohomological object: it acts on $K_T(\mathrm{Hilb}^n) = H^*(B^{\mathrm{ord}})^\vee$, which is the cohomology of the bar complex, not the chain-level complex itself.

(b) The functor α (restriction to associahedra) commutes with passage to cohomology: the $K_n \hookrightarrow \mathrm{FM}_n^{\mathrm{ord}}(\mathbb{C})$ embedding respects the stratification that defines the bar differential.

(c) The functor β (Koszul dualization of deconcatenation) on cohomology is the classical Koszul duality, which is well-defined for any algebra (the coderived correction is invisible at cohomology).

Coassociativity (i) follows from the tautological coassociativity of deconcatenation on T^c, which survives dualization. Quasi-triangularity (ii) follows from the YBE for R^{MO} (proved in [3]).

The explicit formulas (iii) are computed as follows. The Cartan generator $\psi(u)$ is the diagonal part of the R-matrix action on the charge-1 Fock space. On $\mathcal{F}_1 = \bigoplus_\lambda \mathbb{C} \cdot |\lambda\rangle$ (partitions of charge 1, i.e. a single box), the R-matrix eigenvalue is $g(u) = (u-h_1)(u-h_2)(u-h_3)/((u+h_1)(u+h_2)(u+h_3))$. The Gauss decomposition $R = F^{-1} \cdot D \cdot E$ (lower triangular, diagonal, upper triangular) extracts the Drinfeld-type generators. The coproduct is read off from the matrix coproduct $\Delta(T(u)) = T(u) \dot{\otimes} T(u-z)$ applied to the Gauss factors. This is the Schiffmann–Vasserot coproduct for $Y(\widehat{\mathfrak{gl}}_1)$, obtained here from the MO R-matrix via γ^{-1}. □

Remark 6.4 (The class M obstruction). For class L algebras (Kac–Moody), γ^{-1} produces a coproduct at chains: the bar-cobar adjunction is a quasi-isomorphism ($\Omega(B(A)) \xrightarrow{\sim} A$), so all three vertices of the triangle (5) are equivalent at the chain level.

For class M algebras ($\mathcal{W}_{1+\infty}$, Virasoro), the chain-level direct-sum bar-cobar map is *not* a quasi-isomorphism: the nonvanishing shadow coefficient $S_4(\mathrm{Vir}_c) = 10/[c(5c+22)]$ forces a bar cohomology class at weight 4. The triangle (5) holds:

(a) *Cohomology:* the coproduct (7) is well-defined on $H^*(B^{\mathrm{ord}})^\vee = Y(\widehat{\mathfrak{gl}}_1)$.

(b) *Coderived category:* the bar-cobar adjunction $\Omega \circ B \simeq \mathrm{id}$ holds in the coderived sense (Vol I **??**), producing a coproduct on the coderived model of $\mathcal{W}_{1+\infty}$.

(c) *Weight-completed chain level (proved):* the MC5 pattern of Vol I **??** + **??** lifts γ^{-1} to a chain-level coproduct on the weight-completed A_∞-model $\widehat{\mathcal{W}}_{1+\infty}$ (Vol I **??**).

(d) *Direct-sum chain level (genuinely false at generic c).* The weight-4 shadow obstruction rules out a direct-sum chain-level lift; the correct scope is the weight-completed category.

Theorem 6.5 (Chain-level coproduct for class M, weight-completed; (Vol I **??**)). *For $\mathcal{W}_{1+\infty}$ at self-dual level $c = 1$, the cohomological coproduct Δ_z^{MO} of Proposition 6.3 lifts to a chain-level A_∞-coproduct on the weight-completed A_∞-model $\widehat{\mathcal{W}}_{1+\infty}^{A_\infty}$. Making $(\widehat{\mathcal{W}}_{1+\infty}^{A_\infty}, \Delta_z^{A_\infty}, R^{\mathrm{MO}}(z))$ an A_∞-bialgebra in the sense of Markl–Shnider. The Jindal–Kaubrys–Latyntsev vertex coproduct on the CoHA of $\mathbb{C}^3$ [5] provides a chain-level construction for the positive half $Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathrm{CoHA}(\mathbb{C}^3)$; the Drinfeld double extension to the full affine Yangian is absorbed by the weight-completed pattern.*

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